

Operations on RVs

① Expected Value

$$E[X] = \sum_{\text{(discrete)}} x P[X=x] = \int_{-\infty}^{+\infty} x f_X(x) dx \quad \text{(continuous)}$$

② Derived Random Variables

If $Y = g(X)$, Y is a derived random variable.

$$P_Y(y) = \sum_{x: g(x)=y} P_X(x)$$

Expected Value

$$a) E[Y] = \sum_y y P_Y(y)$$

$$b) E[g(X)] = \sum_x g(x) P_X(x)$$

$$c) E[X+Y] = E[X] + E[Y]$$

$$d) E[aX+b] = a E[X] + b$$

} Formulae

$$n=0 : E[X^0] = 1$$

$$n=1 : E[X] = \text{Expected val}$$

$$n=2 : E[X^2] = \text{Mean Square}$$

$$n=3 : E[X^3] = \text{Skewness}$$

$$n=4 : E[X^4] = \text{Kurtosis}$$

③ n-th Moment

$$\hookrightarrow E[X^n] = \sum x^n P_X(x) \quad (\text{discrete})$$

$$E[X^n] = \int_{-\infty}^{+\infty} x^n f_X(x) dx$$

④ Variance and SD

$$a) \text{Var}(X) = E[(X - E[X])^2] = E[(X - \mu)^2] \quad (\text{discrete})$$

$$b) \text{Var}(X) = \int_{-\infty}^{+\infty} (x - \mu)^2 f_X(x) dx \quad (\text{continuous})$$

$$c) \text{Var}[X] = \sum_x (x - \mu_x)^2 P_X(x)$$

$$= \sum_x (x - \mu_x)^2 \sum_y P_{xy}(x, y)$$

$$= \sum_x \sum_y \cancel{P_{xy}} (x - \mu_x)^2 P_{xy}(x, y)$$

$$d) \text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$e) \text{SD}^{(n)} = \sqrt{\text{Var}(X)}$$

⑤ n-th Central Moment

$$E[(x-\mu)^n]$$

$$E[(x-\mu)^n] = \sum_1^n (x-\mu)^n P_x(x) \quad (\text{disc})$$

$$E[(x-\mu)^n] = \int_{-\infty}^{\infty} (x-\mu)^n \cdot f_x(x) dx \quad (\text{cont.})$$

$$n=0 \quad E[(x-\mu)^0] = 1$$

$$n=1 \quad E[(x-\mu)] = \text{Var}[x] = 0$$

$$n=2 \quad E[(x-\mu)^2] = \text{Skewness Var}[x]$$

$$n=3 \quad E[(x-\mu)^3] = \text{Kurtosis Skewness}$$

⑥ Expected value of Two RVs

$$\text{If } W = g(X, Y),$$

$$E(W) = E[g(X, Y)] = \sum_x \sum_y g(x, y) P_{xy}(x, y) \quad (\text{disc.})$$

$$E(W) = E[g(X, Y)] = \int \int g(x, y) f_{x, y}(x, y) dx dy$$

⑦ Covariance

Tells how two RVs change against each other.

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] \quad \begin{array}{l} \rightarrow > 0 \rightarrow \text{positive} \\ \rightarrow < 0 \rightarrow \text{negative} \\ \rightarrow = 0 \rightarrow \text{no linear relation} \end{array}$$

$$\text{Var}[ax + bY] = a^2 \text{Var}[X] + b^2 \text{Var}[Y] + 2ab \text{Cov}(X, Y)$$

⑧ Correlation

Correlation Coefficient : normalized covariance.

$$\rho_{X,Y} = \frac{\text{Cov}[X, Y]}{\sqrt{\text{Var}[X] \text{Var}[Y]}} = \frac{\text{Cov}[X, Y]}{\sigma_X \sigma_Y}$$